Unit 8.01: PCA & Dimensionality Reduction

Q1. Principal Component Analysis PCA projects a dataset from mathbb{R}^{30} onto its top 2 principal components. What is the dimension of each transformed data point?

[ ] A: 15
[ ] B: 30
[ ] C: 28
[x] D: 2

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 2 , so the reduced dimension is 2 .

Q2. Principal Component Analysis PCA projects a dataset from mathbb{R}^{40} onto its top 4 principal components. What is the dimension of each transformed data point?

[ ] A: 20
[ ] B: 40
[ ] C: 36
[x] D: 4

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 4 , so the reduced dimension is 4 .

Q3. Principal Component Analysis PCA projects a dataset from mathbb{R}^{50} onto its top 6 principal components. What is the dimension of each transformed data point?

[ ] A: 25
[ ] B: 50
[ ] C: 44
[x] D: 6

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 6 , so the reduced dimension is 6 .

Q4. Principal Component Analysis PCA projects a dataset from mathbb{R}^{60} onto its top 8 principal components. What is the dimension of each transformed data point?

[ ] A: 30
[ ] B: 60
[ ] C: 52
[x] D: 8

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 8 , so the reduced dimension is 8 .

Q5. Principal Component Analysis PCA projects a dataset from mathbb{R}^{70} onto its top 10 principal components. What is the dimension of each transformed data point?

[x] A: 10
[ ] B: 70
[ ] C: 60
[ ] D: 35

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 10 , so the reduced dimension is 10 .

Q6. Principal Component Analysis PCA projects a dataset from mathbb{R}^{80} onto its top 12 principal components. What is the dimension of each transformed data point?

[x] A: 12
[ ] B: 80
[ ] C: 68
[ ] D: 40

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 12 , so the reduced dimension is 12 .

Unit 8.01: PCA & Dimensionality Reduction

Q7. Principal Component Analysis PCA projects a dataset from mathbb{R}^{90} onto its top 14 principal components. What is the dimension of each transformed data point?

[x] A: 14
[ ] B: 90
[ ] C: 76
[ ] D: 45

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 14 , so the reduced dimension is 14 .

Q8. Principal Component Analysis PCA projects a dataset from mathbb{R}^{100} onto its top 16 principal components. What is the dimension of each transformed data point?

[ ] A: 100
[x] B: 16
[ ] C: 84
[ ] D: 50

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 16 , so the reduced dimension is 16 .